

RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

1. Introduction

3 hrs · asked in **21 of 24 papers** · 3 syllabus sub-topics · core foundational chapter → qualitative comparisons, equivalent circuits, frequency tables, and system protocols.

READ THIS FIRST

This chapter provides guaranteed theory marks in almost every paper (typically 8–14 marks). The examiner repeatedly asks three core questions:

1. **Conventional low frequency vs RF/microwave circuit behavior** (asked in 21 papers) → contrast lumped vs distributed analysis ($d \ll \lambda$ vs $d \gtrsim 0.01\lambda$), breakdown of Kirchhoff's laws, skin effect, and equivalent circuit parasitics of R , L , C .
 2. **Detailed real-world protocols and specifications of microwave communication systems** (asked in 19 papers) → prepare full parametric tables for GSM 900/1800, Satellite C/Ku-band, Terrestrial Microwave link, and WiFi 802.11.
 3. **IEEE microwave frequency band classification** (asked in 6 papers) → memorise the letter-code frequency boundaries (L, S, C, X, Ku, K, Ka, mmWave) and their primary application.
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1.1 Standard Frequency Bands and Features

How is microwave frequency band classified by IEEE? Enumerate features, advantages and disadvantages of microwaves 2+4 3+3 4+4 5+5

HOT 6/24 70 Bh, 70 Ma, 76 Bh, 80 Ba, 79 Bh, 82 Ba, 69 Bh · standard frequency classification

Microwave definition and spectrum

- **Frequency range:** 300 MHz to 300 GHz.
- **Wavelength range:** $\lambda = c/f = 1$ m down to 1 mm (free space).
- **Millimeter waves (mmWave):** 30 GHz to 300 GHz ($\lambda = 10$ mm to 1 mm).
- **Sub-millimeter / Terahertz:** 300 GHz to 3 THz ($\lambda < 1$ mm).

IEEE Radar / Microwave Band Designations

Band	Frequency	Primary Uses
UHF	300–1000 MHz	TV, GSM, GPS, RFID
L	1–2 GHz	Mobile (GSM/LTE), GPS, ATC radar
S	2–4 GHz	Wi-Fi (2.4 GHz), Bluetooth, MW oven
C	4–8 GHz	Satellite TV, weather radar
X	8–12 GHz	Military radar, marine radar, deep space
Ku	12–18 GHz	Direct Broadcast Satellite (DBS TV)
K	18–27 GHz	Police speed radar, water vapor absorption
Ka	27–40 GHz	High-throughput satellite, 5G NR
V	40–75 GHz	60 GHz unlicensed WiGig, inter-sat
W	75–110 GHz	Automotive radar (77 GHz), imaging
mm	110–300 GHz	Radio astronomy, security imaging

Advantages of microwaves

- **High bandwidth:**
 - Bandwidth is 1–10% of carrier frequency.
 - Carrier at 10 GHz offers 1 GHz bandwidth → accommodates thousands of high-speed data channels.
- **High directivity and gain:**
 - Antenna gain $G \propto (D/\lambda)^2$.
 - Small physical antenna diameter D produces sharp, pencil-beam radiation → high EIRP, low interference.
- **Compact physical size:**
 - Component dimensions scale with λ ($\lambda/2$, $\lambda/4$).
 - RF circuits fit on compact MMICs and microstrip boards.
- **Ionospheric transparency:**
 - Waves above 100 MHz penetrate ionosphere without total reflection → enables satellite and space communication.
- **Unique molecular resonance:**
 - Water molecules absorb 2.45 GHz → efficient dielectric heating (cooking, industrial drying).

Disadvantages of microwaves

- **Atmospheric attenuation:**
 - Rain fade and gaseous absorption (H_2O at 22.2 GHz, O_2 at 60 GHz) degrade signals over long paths.
- **Line of sight (LOS) required:**
 - Cannot bend around obstacles (diffraction is minimal); shadows and terrain block transmission.
- **High manufacturing cost:**
 - Tight mechanical tolerances ($< \mu\text{m}$); requires GaAs, InP, or GaN instead of cheap silicon.
- **Complex circuit analysis:**
 - Lumped analysis fails; distributed transmission lines, scattering parameters, and electromagnetic field equations required.

1.2 Behaviour of Circuits: Conventional vs RF/Microwave Bands

Compare the behavior of microwave circuits against conventional low frequency circuits. Differentiate lumped vs distributed analysis 4 5 6 8 10

TOP 21/24 70 Bh, 70 Ma, 71 Ma, 72 Ash, 72 Ma, 73 Bh, 73 Ma, 75 Bh, 76 Bh, 77 Ch, 78 Ch, 79 Ch, 80 Ba, 80 Bh, 80 Ch, 81 Bh, 81 Ba, 82 Bh, 82 Ba, 69 Bh · top repeated topic

- The fundamental governing criterion: d/λ
- Let d = maximum physical dimension of the circuit.
 - Let $\lambda = c/f$ = operating wavelength.
 - **Low frequency** ($d \ll \lambda$, typically $d < 0.01\lambda$):
 - Propagation time of EM wave across the circuit is negligible compared to wave period $T = 1/f$.
 - Phase change across any conductor:
 $\Delta\theta = \beta d = \frac{2\pi d}{\lambda} \approx 0$.
 - Voltage is instantaneous and uniform along an ideal node.
 - **Kirchhoff's Laws (KVL, KCL) hold strictly.**
 - **RF / Microwave** ($d \gtrsim 0.01\lambda$ to $d \approx \lambda$):
 - Propagation delay cannot be ignored; wave travel causes finite phase shift along conductors.
 - Voltage and current become functions of **both position z and time t** : $v(z, t), i(z, t)$.
 - Reflections arise at every impedance discontinuity.
 - Wires exhibit distributed inductance and capacitance; open lines radiate energy like antennas.
 - **Distributed transmission line theory** and Maxwell's equations replace KVL/KCL.

Comparison: Low Frequency vs RF/Microwave Systems

Parameter	Low Frequency / Conventional	RF / Microwave
Circuit theory	Lumped elements (R, L, C discrete blocks).	Distributed elements (R', L', G', C' per meter).
Laws applied	KVL, KCL, Ohm's law.	Maxwell's equations, Telegrapher's wave equations.
Variables	Absolute voltage V , current I .	Incident / reflected wave amplitudes (a, b), S-parameters, power.
Lines used	Wires, twisted pair, PCB traces.	Coaxial cables, Waveguides, Microstrip, Stripline.
Interconnects	Ideal short circuits (zero phase delay).	Transmission lines with characteristic impedance Z_0 and phase delay βz .
Discontinuities	No reflection; current is continuous.	Causes wave reflection, standing waves ($VSWR > 1$).
Radiation	Negligible; energy guided in wire.	Significant radiation if unshielded ($d \approx \lambda/4$).
Tuning	Variable lumped capacitors/inductors.	Reactive stubs, slides, screws, waveguide plungers.
Active devices	BJTs, Op-Amps, Si MOSFETs.	GaAs MESFETs, HEMTs, Gunn diodes, Klystrons, Magnetrons, TWTs.
Measurement	Voltmeter, ammeter, oscilloscope.	Power meter, Bolometer, VSWR meter, Spectrum/Vector Network Analyzer.

Why are S-parameters used in microwave network analysis instead of Z, Y, or h-parameters? 3 4 5

TOP 15/24 70 Bh, 70 Ma, 71 Ma, 72 Ash, 73 Bh, 73 Ma, 74 Ma, 78 Ch, 80 Ba, 80 Ch, 79 Bh, 81 Bh · core justification

Limitations of H, Z, Y parameters at microwave bands

- **Requires open and short circuits:**
 - Z-parameters require open circuits ($I = 0$):
 $Z_{11} = V_1/I_1|_{I_2=0}$.
 - Y-parameters require short circuits ($V = 0$):
 $Y_{11} = I_1/V_1|_{V_2=0}$.
 - h-parameters require both short and open circuits.
- **Ideal opens and shorts are physically impossible:**
 - An open-circuited line at microwave frequencies radiates energy like an open-ended antenna → radiation loss causes $I \neq 0$.
 - Stray fringe capacitance exists at open ends.
 - A short circuit has finite lead inductance ($L_{\text{lead}} > 0$)
→ $\omega L \gg 0$, not a true short ($V \neq 0$).
- **Active device instability:**
 - Transistors (MESFETs, BJTs) frequently become conditionally unstable and oscillate violently under pure open/short terminations.
- **Direct voltage/current measurement impossible:**
 - Probes perturb fields; standing waves make V and I position-dependent.

Why S-parameters succeed

- **Terminated in matched characteristic impedance Z_0 :**
 - S-parameter measurement terminates ports in matched loads ($Z_L = Z_0 = 50 \Omega$).
 - Matched load absorbs all incident energy → zero reflection ($a_2 = 0$).
 - Precision 50Ω broadband loads are easy to manufacture.
- **No parasitic radiation or lead reactance:**
 - Complete absorption eliminates open-circuit radiation and short-circuit inductive ringing.
- **Active devices remain stable:**
 - Resistive 50Ω loading prevents self-oscillation of active transistors during testing.
- **Based on traveling power waves:**
 - Relies on incident wave a and reflected wave b :

$$a = \frac{V + Z_0 I}{2\sqrt{Z_0}}, \quad b = \frac{V - Z_0 I}{2\sqrt{Z_0}}$$

- Directly measurable using directional couplers and Vector Network Analyzers (VNA).

1.3 High-Frequency Parasitics in Passive Components (R, L, C)

Explain the characteristic behavior of passive components (resistor, inductor, capacitor) at RF/microwave frequencies 4 6 8

HOT 5/24 70 Bh, 77 Ch, 79 Ch, 80 Bh, 82 Bh · equivalent circuits and frequency responses

1. Resistors at RF/Microwave

- **Parasitic effects:**
 - **Lead inductance L :** metal leads introduce serial inductance ($L \approx 1\text{--}5 \text{ nH}$).
 - **Stray capacitance C_a :** capacitive coupling between resistor terminals.
 - **Skin effect:** current concentrates in outer conductor layer ($\delta = 1/\sqrt{\pi f \mu \sigma}$) → cross-sectional area decreases → AC resistance increases: $R_{ac} \propto \sqrt{f}$.
- **Equivalent circuit:**
 - Resistor R in series with lead inductance L , shunted by terminal capacitance C_a .
 - At low freq: $Z \approx R$.
 - At medium freq: L causes inductive impedance rise.
 - At high freq: shunt C_a bypasses R , causing impedance to drop.

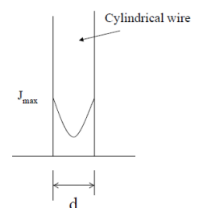
Resistor equivalent model

Resistance

AC → B-fields → Force on charge carriers

Skin depth

$$\delta = \sqrt{\frac{1}{\pi f \mu \sigma}}$$



Equivalent circuit: ideal R with series lead L and parallel parasitic C_a .

2. Inductors at RF/Microwave

– Parasitic effects:

- **Winding resistance** R_s : coil wire resistance, increased by skin and proximity effects ($R_{ac} \propto \sqrt{f}$).
- **Inter-turn capacitance** C_d : electric field lines between adjacent coil turns create distributed parallel capacitance.

– Self-Resonant Frequency (SRF), f_{SR} :

$$f_{SR} = \frac{1}{2\pi\sqrt{LC_d}}$$

– Behavioral regions:

- $f < f_{SR}$: inductive reactance dominates ($\omega L > 1/\omega C_d$); X_L increases with f .
- $f = f_{SR}$: parallel resonance $\rightarrow$ impedance peaks to a very large purely resistive value; quality factor $Q \rightarrow 0$.
- $f > f_{SR}$: **Inductor acts as a capacitor!** Inter-turn capacitance bypasses the coil ($\omega C_d > 1/\omega L$).

3. Capacitors at RF/Microwave

– Parasitic effects:

- **Lead inductance** L_s (**ESL**): connecting leads and plates introduce series inductance ($L_s \approx 1\text{--}2$ nH).
- **Dielectric loss** R_p : leakage resistance across dielectric ($\tan \delta = \epsilon''/\epsilon'$).
- **Equivalent series resistance** R_s (**ESR**): resistance of plates and leads.

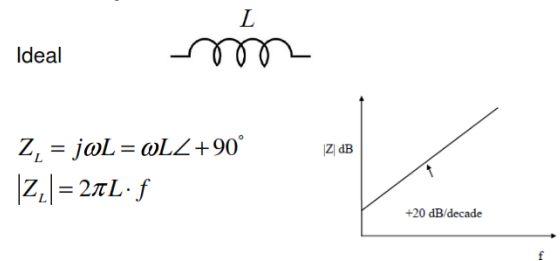
– Series Self-Resonant Frequency, f_{SR} :

$$f_{SR} = \frac{1}{2\pi\sqrt{L_s C}}$$

– Behavioral regions:

- $f < f_{SR}$: capacitive reactance dominates ($1/\omega C > \omega L_s$); impedance drops with f .
- $f = f_{SR}$: series resonance $\rightarrow$ impedance reaches a minimum equal to ESR (R_s); optimal for bypass/decoupling.
- $f > f_{SR}$: **Capacitor acts as an inductor!** ESL dominates ($\omega L_s > 1/\omega C$); impedance rises with f .

Inductor equivalent model

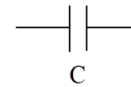


Inductor equivalent circuit: inductance L with series loss R_s shunted by distributed winding capacitance C_d .

Capacitor equivalent model

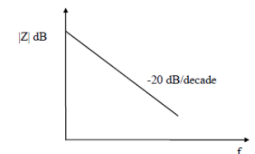
Capacitors

Ideal



$$Z(j\omega) = \frac{1}{j\omega C} = \frac{1}{\omega C} \angle -90^\circ$$

$$|Z| = \frac{1}{2\pi C} \cdot \frac{1}{f}$$



Capacitor equivalent circuit: ideal C with series lead inductance L_s , series resistance R_s , and dielectric leakage R_p .

1.4 Comparison with Other Wave Types

Compare and contrast microwave signals with acoustic and seismic waves 45

HOT 3/24 72 Ma, 72 Ash, 75 Bh · physical wave mechanisms

Physical wave nature

- **Microwave signals:**
 - Electromagnetic transverse waves ($\vec{E} \perp \vec{H} \perp \vec{k}$).
 - Propagates in vacuum (no medium required).
 - Propagation speed: $c \approx 3 \times 10^8$ m/s.
 - Frequencies: 300 MHz to 300 GHz; wavelengths: 1 m to 1 mm.
- **Acoustic (sound) waves:**
 - Mechanical longitudinal pressure waves (compression / rarefaction).
 - Requires material medium (gas, liquid, solid); zero propagation in vacuum.
 - Speed: ~ 343 m/s in air, ~ 1500 m/s in water.
 - Frequencies: 20 Hz to 20 kHz (ultrasound > 20 kHz).
- **Seismic waves:**
 - Mechanical waves traveling through Earth's geological strata.
 - Body waves (P -waves longitudinal, S -waves transverse shear) + Surface waves (Rayleigh, Love).
 - Speed: 2–8 km/s depending on rock density.
 - Infrasonic frequencies: 0.01 Hz to 50 Hz.

Comparison Table

Property	Microwaves	Acoustic	Seismic
Wave type	EM (Transverse)	Mechanical (Longitudinal)	Mechanical (Long. & Shear)
Medium	Not required (vacuum)	Mandatory (gas/liq/solid)	Solid earth / rock
Velocity	3×10^8 m/s	~ 343 m/s (air)	2000–8000 m/s
Frequency	10^8 – 10^{11} Hz	20–20000 Hz	0.01–50 Hz
Wavelength	1 mm – 1 m	17 mm – 17 m	100 m – 100 km
Information	Massive data rate (Gbps)	Audio/speech, sonar	Geological fault analysis
Absorption	Rain / moisture fade	Viscous fluid damping	Geometric spreading, rock damping
Application	Radar, satellite, 5G	Audio, ultrasound NDT	Earthquake detection, oil survey

1.5 Microwave Applications and Real-World System Protocols

Prepare the detail real-world protocols and specifications of any 3 (or 5) major microwave communication systems 4+6 2+6 4+4 8 10

TOP 19/24 70 Bh, 70 Ma, 73 Bh, 76 Bh, 77 Ch, 78 Ch, 79 Bh, 80 Bh, 81 Ba, 81 Bh, 82 Bh, 82 Ba · top repeated exam requirement

General microwave communication system architecture

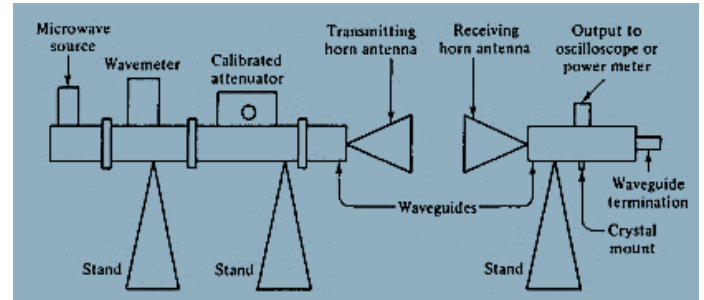
- **Transmitter (Tx):** Baseband source → Modulator (QPSK/QAM) → Up-converter (Mixer + Local Oscillator) → Bandpass Filter → High Power Amplifier (HPA: TWT or GaN SSPA) → Duplexer/Circulator → Directional Antenna.
- **Channel:** Free space line-of-sight (LOS) path with atmospheric attenuation and multipath fading.
- **Receiver (Rx):** Directive Antenna → Duplexer → Low Noise Amplifier (LNA) → Down-converter (Mixer + LO) → IF Filter / Amplifier → Demodulator → Baseband output.

Major application areas

- **Telecommunications:** Cellular BTS, satellite, microwave LOS trunk radio.
- **Radar & Navigation:** Air traffic control (ASR), weather radar, GPS/GLONASS.
- **Industrial & Domestic:** Microwave ovens (2.45 GHz), rubber vulcanization, food drying.
- **Medical:** Hyperthermia cancer treatment, microwave diathermy, tomography.
- **Scientific:** Radio astronomy, remote sensing, atmospheric sounding.

Detailed Specifications and Protocols of Five Major Microwave Communication Systems

Microwave system block diagram



Block diagram of general microwave communication link: Tx upconversion, HPA, antenna, propagation, LNA, downconversion, and detector.

Parameter	1. GSM 900 / 1800	2. Satellite (Ku-band)	3. Terrestrial MW Link	4. Wi-Fi (802.11ac)	5. Airport Radar (ASR)
Operating band	UHF / L-band (900/1800 MHz)	Ku-band (12–18 GHz)	C / X-band (6–11 GHz)	S / C-band (2.4 & 5 GHz)	S-band (2.7–2.9 GHz)
Uplink / Tx Freq	890–915 MHz (GSM 900)	14.0–14.5 GHz	5.925–6.425 GHz	5.15–5.85 GHz	2.7–2.9 GHz pulsed
Downlink / Rx Freq	935–960 MHz (GSM 900)	11.7–12.2 GHz	3.700–4.200 GHz	5.15–5.85 GHz	Reflected echo
Channel bandwidth	200 kHz per carrier	36–54 MHz transponder	28–56 MHz per channel	20, 40, 80, 160 MHz	1–5 MHz chirp
Duplexing mode	FDD (45 MHz duplex split)	FDD (2–2.5 GHz split)	FDD (frequency separation)	TDD (CSMA/CA)	Monostatic simplex (pulsed)
Multiple access	TDMA (8 time slots / carrier)	FDMA / TDMA / DVB-S2	Point-to-Point dedicated	CSMA/CA, OFDMA	Pulse timing / PRF
Modulation scheme	GMSK ($BT = 0.3$)	QPSK, 8PSK, 16-APSK	64-QAM, 256-QAM, 1024-QAM	OFDM w/ 256-QAM	Linear FM Chirp / Barker
Data rate	270.833 kbps gross	30–90 Mbps per transponder	Up to 1 Gbps per link	Up to 1.3 Gbps	Target range & velocity
Tx Output Power	20–40 W (+43–46 dBm)	50–150 W TWTA	1–10 W (+30–40 dBm)	100–200 mW (+20 dBm)	1–25 kW peak
Antenna type	Sectorized panel (120°)	Parabolic reflector (1.2–3.0 m)	Parabolic dish w/ radome	Printed patch array / dipole	Parabolic cosecant-squared
Antenna gain	15–18 dBi	38–45 dBi	35–44 dBi	3–6 dBi	32–35 dBi
Coverage / Range	1–35 km cell radius	36000 km (GEO orbit)	30–60 km per hop	30–100 m indoor	60–80 NM (100–150 km)

1.6 Microstrip Lines vs Strip-Lines

Differentiate between microstrip lines and strip-lines. Discuss their transmission properties 5

PIN 1/24 71 Ma · planar transmission lines

Microstrip Line

- **Geometry:**
 - Conducting strip of width W on top of dielectric substrate (h, ϵ_r).
 - Ground plane on the bottom; upper surface exposed to air (ϵ_0).
- **Wave mode:**
 - **Quasi-TEM mode:** fields exist partly in dielectric (ϵ_r) and partly in air ($\epsilon_0 = 1$).
 - Phase velocities in the two media differ → pure TEM cannot be supported.
 - Characterized by effective dielectric constant ϵ_{eff} :

$$1 < \epsilon_{\text{eff}} < \epsilon_r, \quad v_p = \frac{c}{\sqrt{\epsilon_{\text{eff}}}}$$

- **Dispersion:** Moderate dispersion at higher frequencies as fields pull into substrate.
- **Fabrication:** Very easy; standard photolithography; surface accessible for discrete component mounting (SMD, diodes, FETs).
- **Radiation:** Unshielded; radiates at bends and discontinuities.

Strip-line

- **Geometry:**
 - Conducting strip embedded symmetrically inside dielectric medium between **two parallel ground planes**.
- **Wave mode:**
 - **Pure TEM mode:** dielectric is completely homogeneous around the conductor.
 - Phase velocity is frequency-independent:

$$v_p = \frac{c}{\sqrt{\epsilon_r}}$$

- **Dispersion:** Non-dispersive; wide broadband operation.
- **Shielding:** Completely enclosed between two ground planes → **zero radiation loss**, no crosstalk with adjacent circuits.
- **Fabrication:** Difficult; multi-layer sandwich lamination; impossible to mount surface components after manufacturing.
- **Applications:** High-performance broadband passive circuits (couplers, filters, power dividers).

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.